

EduGenie: Google Gemini Powered Learning Assistant

Phase 2

ARCHITECTURAL DIAGRAM

Pre-requisites

1. Python 3.10+

- Official Documentation: [Google AI for Developers](#)
- Installation Guide: [Tom's Hardware](#)
- Popular Tutorial: [Python Programming Tutorial - Full Course for Beginners](#)
- Installation Steps:

1. Download the latest Python 3.10+ installer for your operating system from the official website.
2. Run the installer and ensure you check the box that says "Add Python to PATH".
3. Follow the installation prompts to complete the setup.
4. Verify the installation by opening a terminal or command prompt and typing `python --version`.

2. FastAPI Framework –

- Official Documentation: [FastAPI](#)
- User Guide: [FastAPI](#)
- Popular Tutorial: [FastAPI Crash Course](#)

3. HTML & CSS – Basic templating used in `/templates` and `/static`

4. Google Gemini API Key

- Official Documentation: [Google AI for Developers](#)
- Setup Guide: [GeeksforGeeks](#)
- Popular Tutorial: [How to Use Google Gemini API Key](#)
- Setup Steps:

1. Visit the Google AI Studio and sign in with your Google account.
2. Create a new project and enable the Gemini API.
3. Generate an API key and securely store it.

5. Uvicorn (ASGI Server)

- Official Documentation: [PyPI](#)
- Installation Steps:

1. Install Uvicorn using pip: `pip install uvicorn`

6. Jinja2 (HTML Templating Engine)

- Official Documentation: [Jinja2 Documentation](#)
- Installation Steps:

1. Install Jinja2 using pip: `pip install jinja2`